

Research paper

The association between N-acetyl-L-cysteine and development of depression: A longitudinal cohort study in all patients with chronic obstructive pulmonary disease in Region Stockholm

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ABSTRACT

Background: Depression is a major cause of illness and sick leave and has a prevalence of 3,4% globally. Approximately 30–55% of patients suffering from depression do not respond to existing treatments. N-Acetyl-L-cysteine (NAC) is currently used as mucolytic agent and as antidote in paracetamol overdose. Lately, NAC has generated interest as neuroprotective agent. It is not known whether NAC can prevent depression.

Research question: This study investigated whether higher consumption of NAC is associated with a lower risk of developing depression in patients with Chronic Obstructive Pulmonary Disease (COPD).

Study design and method: The longitudinal cohort study included patients over 40 years old in Stockholm with COPD and/or chronic bronchitis diagnosis 2014–2016 without depression 2014–2019 ($n = 16,875$). Exposure consisted of defined daily doses (DDD) of NAC 2014–2019, and the outcome was occurrence of depression diagnosis 2020–2024.

Result: Cox regression adjusted for age and sex revealed a hazard ratio for depression of 1.37 (95% CI 1.25–1.52) with low NAC exposure (41–875 DDD) and of 1.29 (95% CI 1.09–1.52) with high exposure (>875 DDD). The associations were somewhat attenuated when adjusted for hospital admissions and other COPD medications.

Conclusion: Contrary to our hypothesis, our results show an increased hazard ratio of depression in patients treated with NAC. This most likely reflects the strong association, shown in other studies, between more severe COPD disease and an increased risk of depression. In this scenario, DDD of NAC acts as a proxy for disease severity.

1. Introduction

Chronic obstructive pulmonary disease (COPD) is a common disease with a global prevalence estimate of 10,3% (https://goldcopd.org/wp-content/uploads/2024/11/GOLD-2025-Report-v1.0-15Nov2024_WMV.pdf, 2025; Adelooye et al., 2022). Patients with COPD are at higher risk of developing depression compared to the general population (Stållberg et al., 2018). Moreover, depression comorbidity in COPD is associated with poorer outcome in terms of higher mortality risk, longer hospitalizations (Ng et al., 2007) and lower health related quality of life (Blakemore et al., 2014). Mechanics underlying the association between

depression in COPD are not yet fully understood. Risk factors are most likely multifactorial and include age, female sex, smoking and COPD disease severity (Weiss et al., 2023). Lower socioeconomic status may also be a link between COPD and depression as it is a risk factor for both conditions (Gershon et al., 2012; Domènech-Abella et al., 2018). Successfully treating depression in patients with COPD may improve overall quality of life and reduce hospitalization rates in this large and exposed patient group (Rahi et al., 2023).

Traditionally, patients with COPD have been treated with N-acetyl-L-cysteine (NAC) for its mucolytic properties. NAC has a long history, dating back to the 1960s, of clinical use as a mucolytic agent and as an

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antidote for paracetamol overdose (Raghu et al., 2021), and lately there is growing interest in using NAC to treat neuropsychiatric conditions, e. g. depression, obsessive compulsive disorders and autism (Bradlow et al., 2022). NAC is a synthetic derivative of the endogenous amino acid L-cysteine and a precursor to tripeptide glutathione (Raghu et al., 2021). While it is still not known exactly how NAC can be beneficial in treating neuropsychiatric conditions, several possible mechanisms have been described in scientific literature. These include regulation of several neurotransmitters, regulation of inflammatory mediators, and regulation of oxidative homeostasis (Smaga et al., 2021; Barlattani et al., 2025). The path that draws most attention is the positive impact NAC has on regulation of oxidative homeostasis, thus reducing oxidative stress. Reactive Oxygen Species (ROS) are used by human cells for cell signaling as well as defense against invasive microorganisms (Bhatt et al., 2020), but the level of ROS must be carefully balanced with antioxidative mechanisms to achieve redox homeostasis or proinflammatory mediators will lead to oxidative stress and cell necrosis (Bhatt et al., 2020). The brain is more susceptible to damage from increased levels of ROS than any other organ in the body as it is a major consumer of oxygen and its high levels of oxidizable lipids (Bhatt et al., 2020). Lately, it has been shown that oxidative stress in the brain can be one of the mechanisms behind development of depression (Bhatt et al., 2020). However, while NAC can reduce oxidative stress its effect on depression is still not proven.

NAC is not currently prescribed for depression treatment, but there is some evidence that it may ameliorate depressive symptoms (Yang et al., 2018; Song et al., 2019; Wang et al., 2024; Fernandes and Gupta, 2019; Hasebe et al., 2017; Berk et al., 2014; Porcu et al., 2018; Peng et al., 2024). Pre-clinical studies show that NAC can attenuate anti-depressant-like effects in rats (Yang et al., 2018; Song et al., 2019; Wang et al., 2024). One such study showed that exposing rats for chronic unpredictable mild stress for three weeks introduced depressive-like symptoms which were reversed by one week NAC treatment (Fernandes and Gupta, 2019). Interestingly, this effect was dose related: no significant effect could be seen with NAC 25 mg/kg, whereas 50 mg/kg and 100 mg/kg gave effect. In contrast, clinical studies on the efficacy of NAC on depression in humans have shown mixed results (Hasebe et al., 2017; Berk et al., 2014; Porcu et al., 2018). However, a recent systematic review and meta-analysis by Peng et al. indicates that NAC is more effective than placebo in alleviating depressive symptoms (Peng et al., 2024). The same meta-analysis suggests that the optimal dose for effectively alleviating depressive symptoms ranges from 1000 to 2750 mg per day, yet it showed no significant effect related to treatment duration (Peng et al., 2024). To the best of our knowledge there are no published clinical studies that have investigated a possible preventive effect of NAC on the development of depression.

While recognizing that the association may be challenging to prove and that the dosage may not be sufficient to have an antidepressant effect, we hypothesize that higher consumption of NAC is associated with a lower risk of developing depression in patients with COPD. The aim of this study was to examine whether patients in Region Stockholm, diagnosed with either COPD or chronic bronchitis, have a lower risk of being diagnosed with depression if they were treated with NAC compared to non-users.

2. Materials and methods

This study was a longitudinal cohort study based on eleven years of data collected from healthcare registers in Region Stockholm (2014–2024).

2.1. Study population and data source

The study population included all patients living in Stockholm with at least one physician's visit between 2014 and 2016, during which they were diagnosed with either chronic obstructive pulmonary disease,

chronic bronchitis, or both (ICD10-codes J41–J44, J47). Younger patients, born after 1974, were not included, as they are fewer in number and typically have a different etiology than the main patient group with COPD. Patients who were diagnosed with depression between 2014 and 2019 were excluded from the study population (ICD10-codes F32–F33, F41.2). Additionally, patients that collected equal to or more than 200 Defined Daily Doses (DDD) of antidepressants, ATC code N06A*, collected from any pharmacy in Sweden during 2014–2019, were excluded. Patients who either moved in or moved out during the study between 2014 and 2024 or who died between 2014 and 2019 were also excluded from the study population. Patient co-morbidities were based on documented diagnoses during physician's visit between 2014 and 2019. The flowchart of study participants can be found in Fig. 1. Data were sourced from the Stockholm Regional Health Care Data Warehouse (VAL databases) which has been described in detail elsewhere (Carlsson et al., 2013).

2.2. Exposures

The exposure group consists of patients that were treated with NAC, ATC code R05CB01. Exposure categories were defined based on the number of DDD, i.e. 500 mg, of NAC collected from any pharmacy in Sweden between 2014 and 2019. As the study population included individuals diagnosed with either chronic obstructive pulmonary disease, chronic bronchitis, or both during 2014–2016, all individuals had the potential to collect NAC for a minimum of three years. Group 1 was designated as the control group and called “none”, consisting of individuals who had collected up to a maximum 40 DDD of NAC during the 6-year period. Group 2, named “low usage”, contains individuals that have collected between 41 and 875 DDD of NAC, corresponding to up to one dose, i.e. 200 mg, per day during the 6-year period. Group 3, “high usage”, contains individuals that have collected more than 875 DDD of NAC, corresponding to more than one dose, i.e. 200 mg, per day during the 6-year period. Selection of cut-off limits for each of the categories was based on clinical reasoning as there is no previous research to guide the decision. The categories were later verified to ensure that they do not contain too small groups of participants to detect a clinically meaningful effect.

2.3. Outcomes

The ICD-10 codes F32 (depressive episode), F33 (recurrent depressive disorder) and F41.2 (mixed anxiety and depressive disorder) were merged into one binary variable, representing depression diagnosis. Furthermore, collection of equal to or more than 200 DDD of antidepressants, ATC code N06A*, was defined in another binary variable representing subclinical depression. Both variables were merged to define the outcome depression, coded as a binary variable.

2.4. Statistics

Descriptive statistics were used to present relevant characteristics of the cohort participants, and the exposure categories and cox regression, with weeks of follow-up as the underlying timescale, was used to calculate hazard ratios (HR) for depression during 2020–2024 for each of the exposure groups (low, high) compared to the referents (none). Start of follow-up was Jan 1st, 2020. End of follow-up was defined as date of first depression outcome, death or study end (Dec 31st, 2024), whichever came first. Visual representation of the study design can be found in Fig. 2. Additionally, men and women were analyzed separately.

Model 1 was adjusted for age to account for any differences in age distribution between the different exposure categories. Age is a continuous variable, captured for all individuals at the start of study, i.e. 2014. Model 2 was adjusted for age and sex as it is well established that females have a higher risk of developing depression (Lancet Psychiatry, 2022). Sex is a binary variable, either female or male in the register.

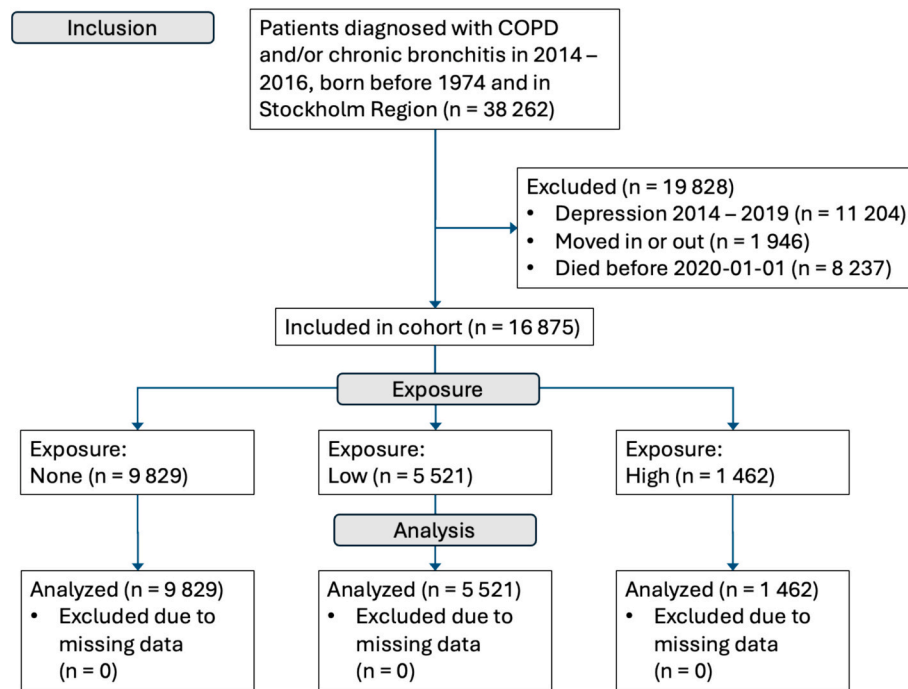


Fig. 1. Study population. Flowchart of the included individuals in the study.

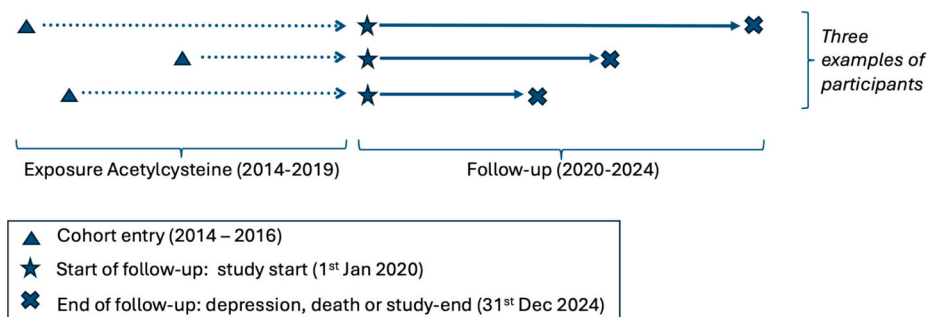


Fig. 2. Study design. Visual representation of study design.

Model 3 was adjusted for age, sex and hospital admissions as a proxy for overall severity of disease and comorbidity. Hospital admissions were defined as a continuous variable, calculated as the number of hospital admissions during the years 2014–2019 irrespective of reason. Model 4 was adjusted for age, sex and COPD medications collected. COPD medications were defined as a continuous variable based on the number of DDD collected between 2014 and 2019 where ATC code starts with either R03A (Adrenergic, inhalants) or R03B (Other drugs for obstructive airway diseases, inhalants). All COPD medications other than NAC were pooled together into one variable. This variable was used as a proxy for severity of COPD disease.

All analyses were conducted using R software, version 4.1.1, packages base, datasets, dplyr, methods, stats, stringr, survival, tidyr and utils.

2.5. Ethics

The study has been approved by the Swedish Ethical Review Authority (Dnr. 2021–01016 with amendments 2021-05735-02, and 2024-05462-02).

3. Results

Characteristics of the cohort population are shown in [Table 1](#). In total, 16875 individuals made up the cohort population. Of these, a larger proportion were female (54.2%) compared to male (45.8%). Most of the cohort had collected no or up to 40 DDD of NAC (category “none” includes 58.6%), a third has collected less than one dose per day (category “low” includes 32.7%), and a small portion has collected more than one dose per day (category “high” includes 8.66%). Comorbidities, which have been shown to correlate with depression diagnoses ([Wallensten et al., 2024](#)), are distributed relatively evenly amongst the exposure category groups, except for anxiety disorders (4.66% for category “none” vs 6.70% for category “high”) and sleep disorders (14.5% for category “none” vs 17.9% for other categories).

The hazard ratio for being diagnosed with depression was increased for all groups treated with NAC compared to those not treated with NAC (Table 2a). When adjusted for age, sex, number of hospitalizations, and COPD medication other than NAC, hazard ratios were attenuated, but still statistically significant for all categories. Analyzing the group of females separately, the hazard ratio follows the main analysis (Table 2b), and when analyzing the group of males separately (Table 2c) the results are similar, but non-significant for the high category.

A sensitivity analysis which introduced a “very high” category

Table 1
Characteristics of cohort participants per exposure category.

Characteristics/Exposure category	All n = 16,875 (100%)	None n = 9892 (58.6%)	Low n = 5521 (32.7%)	High n = 1462 (8.66%)
Mean age, years, at entry (SD)	68.5 (9.53)	67.8 (9.70)	69.4 (9.28)	70.4 (8.78)
Median age, years, at entry	69	69	70	71
Sex, number (%)				
Female	9154 (54.2)	5254 (53.1)	3108 (56.3)	792 (54.2)
Male	7721 (45.8)	4638 (46.9)	2413 (43.7)	670 (45.8)
Average number of hospital visits during 2014–2019 irrespective of reason, number (SD)	2.77 (4.76)	2.31 (4.52)	3.32 (4.88)	3.75 (5.38)
Average number of DDD NAC collected ^a per calendar year 2014–2019, number (SD)	37.3 (75.8)	1.28 (2.47)	46.0 (35.6)	247.4 (90.8)
Average number of DDD non-NAC COPD medications collected ^b per calendar year 2014–2019, number (SD)	336.5 (341.3)	243.5 (267.2)	412.2 (347.6)	679.3 (418.1)
Diagnosed with depression or collected at least 200 DDD of antidepressants ^c 2020–2024, number (%)	1781 (10.6)	926 (9.36)	689 (12.5)	166 (11.4)
Comorbidities [ICD-code], number (%)				
Alcohol related disorders [F10]	870 (5.16)	530 (5.36)	275 (4.98)	65 (4.45)
Substance use disorder [F11-F16, F18, F19]	193 (1.14)	115 (1.16)	64 (1.16)	14 (0.958)
Schizophrenia, schizotypal disorders and delusional disorders [F20-F29]	185 (1.10)	113 (1.14)	58 (1.05)	14 (0.958)
Manic disorders [F30-F31, F34]	87 (0.516)	47 (0.475)	31 (0.561)	9 (0.616)
Anxiety disorders [F40-F42]	925 (5.48)	461 (4.66)	366 (6.63)	98 (6.70)
Sleep disorders [F51, G47]	2682 (15.9)	1430 (14.5)	991 (17.9)	261 (17.9)

^a ATC code R05CB01.
^b ATC codes starting with R03A or R03B.
^c ATC codes starting with N. Abbreviations: ATC, Anatomical Therapeutic Chemical, ICD International Classification of Diseases version 10, SD Standard Deviations.

Table 2a
Cox regression models for the association between treatment with acetylcysteine and the risk of depression 2020–2024 expressed as hazard ratios and 95% confidence intervals for all individuals.

Acetylcysteine category	None	Low	High
Model 1 Depression ^a , HR (95% CI)	1.0	1.38 (1.25–1.53)	1.29 (1.09–1.52)
Model 2 Depression ^a , HR (95% CI)	1.0	1.37 (1.25–1.52)	1.29 (1.09–1.52)
Model 3 Depression ^a , HR (95% CI)	1.0	1.36 (1.23–1.50)	1.26 (1.07–1.49)
Model 4 Depression ^a , HR (95% CI)	1.0	1.35 (1.22–1.50)	1.23 (1.03–1.47)

^a ICD-10 code F32 or F33 or more than 200 DDD of antidepressants collected. Model 1 was adjusted for age. Model 2 was adjusted for age and sex. Model 3 was adjusted for age, sex and number of hospitalizations (2014–2019). Model 4 was adjusted for age, sex and COPD medicine other than NAC (2014–2019). Abbreviations: HR hazard ratio, CI confidence.

Table 2b
Cox models for the association between treatment with acetylcysteine and the risk of depression 2020–2024 expressed as hazard ratios and 95% confidence intervals for all females.

Acetylcysteine category	None	Low	High
Model 1 Depression ^a , HR (95% CI)	1.0	1.43 (1.26–1.62)	1.41 (1.15–1.73)
Model 2 Depression ^a , HR (95% CI)	1.0	1.43 (1.26–1.62)	1.41 (1.15–1.73)
Model 3 Depression ^a , HR (95% CI)	1.0	1.39 (1.23–1.58)	1.34 (1.09–1.65)
Model 4 Depression ^a , HR (95% CI)	1.0	1.39 (1.23–1.58)	1.23 (1.07–1.66)

^a ICD-10 code F32 or F33 or more than 200 DDD of antidepressants collected. Model 1 was adjusted for age. Model 2 was adjusted for age and sex. Model 3 was adjusted for age, sex and number of hospitalizations (2014–2019). Model 4 was adjusted for age, sex and COPD medicine other than NAC (2014–2019). Abbreviations: HR hazard ratio, CI confidence.

Table 2c
Cox models for the association between treatment with acetylcysteine and the risk of depression 2020–2024 expressed as hazard ratios and 95% confidence intervals for all males.

Acetylcysteine category	None	Low	High
Model 1 Depression ^a , HR (95% CI)	1.0	1.30 (1.11–1.53)	1.12 (0.85–1.48)
Model 2 Depression ^a , HR (95% CI)	1.0	1.30 (1.11–1.53)	1.12 (0.85–1.48)
Model 3 Depression ^a , HR (95% CI)	1.0	1.29 (1.10–1.51)	1.11 (0.84–1.46)
Model 4 Depression ^a , HR (95% CI)	1.0	1.29 (1.09–1.53)	1.10 (0.81–1.47)

^a ICD-10 code F32 or F33 or more than 200 DDD of antidepressants collected. Model 1 was adjusted for age. Model 2 was adjusted for age and sex. Model 3 was adjusted for age, sex and number of hospitalizations (2014–2019). Model 4 was adjusted for age, sex and COPD medicine other than NAC (2014–2019). Abbreviations: HR hazard ratio, CI confidence.

revealed hazard ratio for depression of 1.24 (95% CI 0.78–1.98) when adjusted for sex and age (Supplementary Table 1). The very high category ($n = 165$) included patients that had collected more than 2189 DDD of NAC, corresponding to one DDD, i.e. 500 mg, per day during the exposure period.

4. Discussion

Contrary to our hypothesis, our results indicate an increased risk of developing depression in patients with COPD and/or chronic bronchitis who are treated with NAC. The most likely explanation to this finding is confounding by indication, meaning that NAC consumption represents a proxy for disease severity, because patients with more severe disease probably collect higher quantities of NAC and also develop depression to a higher degree. We observed that NAC consumption correlates with other severity proxies, namely hospitalization and the use of other COPD medicine, yet adjusting for those factors did not remove the observed association between NAC and depression.

Similarly to our results, there is a known relationship between COPD, increased COPD severity, and the risk of depression (Schneider et al., 2010; Lee et al., 2013; von Siemens et al., 2019). A recent study

investigated the association between symptom severity and exacerbation risk in patients with COPD and clinically significant anxiety and depression symptoms in 2664 patients with COPD. The study found that patients with more severe symptoms and higher risk for exacerbation had a significantly increased risk for depression compared to patients with less severe symptoms (Weiss et al., 2023). In this study the population were separated into groups as per the Global Initiative for Obstructive Lung Disease (GOLD) disease severity groups as per 2011 standard, where Group A displays low symptoms and low exacerbation risk, Group B displays high symptoms and low exacerbation risk, Group C displays low symptoms and high exacerbation risk, and Group D displays high symptoms and high exacerbation risk (Weiss et al., 2023). The prevalence of clinically significant depression symptoms was 11.8% in group A, 33.3% in group B, 18.6% in group C and 43.0% in group D (Weiss et al., 2023). The strong association between COPD severity and risk of developing depression (Weiss et al., 2023; Schneider et al., 2010; Lee et al., 2013; von Siemens et al., 2019) is likely to overshadow any impact of NAC reducing the risk of developing depression. Unfortunately, in our study, we did not have access to data about the severity of disease as per the GOLD severity groups. Instead, we tested other indicators of disease severity such as hospitalizations and COPD medication other than NAC. However, the results remain despite compensating for those indicators, meaning that they most likely do not fully capture the severity of COPD disease.

In our study we found similar patterns for males and females, but the results differ slightly in hazard ratio and level of significance. When investigating females separately, the hazard ratios for depression outcome in those treated with NAC were elevated, for both low and high exposure categories in each of the four models, compared to when both sexes were investigated together. The only exception was model 3, adjusted for age, sex and hospital admissions, where females had a slightly reduced risk in the low exposure category compared to when both sexes investigated together categories. However, when investigating males separately, the results are not significant for high NAC consumption. There are many possible explanations for the divergent significant levels between females and males. First, the number of patients in the male group is lower than in the female group, which is as expected as the prevalence of COPD is lower amongst males in Sweden (Marshall et al., 2022). Second, the overall number of diagnosed depressions amongst males is also fewer, which also is in-line with expectations as depression is less common amongst males (*Lancet Psychiatry*, 2022). Third, when it comes to severity of disease, as indicated by the standard deviation for average number of hospital admissions (females 4.39 vs men 5.15), the males in the cohort seem to be a more diverse group compared to the females.

Exposure is based on DDD of NAC collected from pharmacies in Sweden. The cut-off for exposure categories has been selected based on clinical and logical reasoning. In a sensitivity analysis we introduced an additional “very high” category consisting of individuals that use at least one DDD per day, i.e. 500 mg, but even that did not change the overall result. In fact, the risk estimates for the very high NAC use were non-significant.

The population in our study, patients with COPD and/or chronic bronchitis, is older and has significantly lower NAC treatment dosage compared to other populations in previous studies of NAC treatment and depression. A randomized controlled trial from 2014 (Berk et al., 2014), which investigated the efficacy of adjunctive NAC in major depressive disorder (MDD), found that the efficacy of NAC separated from placebo in the mid-tertile groups, while the younger and older participants failed to separate. In that study, the mean age was 50.2 with a SD of 12.7 years. The participant group is considerable younger compared to the participants in our cohort (mean age 68.5, SD 9.53), which may explain why we failed to see any positive impact on NAC treatment to prevent depression. Precisely why age has this impact is currently unclear. Similarly, a meta-analysis investigating efficacy of NAC for patients with depression (Peng et al., 2024) shows that the efficacy of NAC is dose

dependent and that the daily dose of 1000 mg to 2750 mg is needed to alleviating depressive symptoms. Even the lowest point of this dose range is almost twice the daily recommended dose of NAC for patients with COPD, which is up to 600 mg. The lower NAC dosage could be one potential reason for NAC not preventing development of depression in patient group with COPD. A future randomized controlled trial with a daily dose range as indicated by the meta-analysis mentioned above could clarify if NAC has the potential to prevent development of depression in this group.

Strengths of the study are the long duration of the follow-up, the size of the cohort and the completeness of data. We also do not rely on any self-assessment, but rather on licensed physicians' documented diagnosis decisions. Due to the nature of data, no imputations have been necessary, except for 72 individuals where only year or month of death were known and where first day of year and/or month were used as approximation.

A limitation is that diagnoses in this large clinical dataset are recorded according to clinical practice and knowledge support systems that do not necessarily match the inclusion criteria typically used in clinical studies on depression. This can result in both under- and over-estimation of prevalence and it not possible to know how this affects the result.

Furthermore, it is well established that there is an increased risk of repeated episodes of depression compared to having a first episode of depression (Eaton et al., 2008; Olaya et al., 2025). Our study design ensures that participants did not have a depression diagnosis during 2014–2019, but it is not known whether they had a diagnosis prior to 2014. Additionally, it is a known fact that depressive symptoms and/or depression are common amongst the older population, yet symptoms are often overlooked, and depression is underdiagnosed especially amongst older individuals (Borglin et al., 2019; Bacigalupe et al., 2024). To reduce the risk of missing subclinical cases or inconsistent clinical documentation, regular collection of antidepressants was used as an alternative criterion for depression. However, this introduces the limitation that patients prescribed antidepressants for other reasons, e.g. anxiety disorders or neuropathic pain, are incorrectly classified as having the outcome depression (Deferio et al., 2018). Overall, we believe that this is more relevant than to only include registered depression diagnoses. A clear need for treatment is now captured in our outcome.

A limitation is that this study does not have access to validated disease severity scores, such as GOLD classification, spirometry data nor exacerbation frequency for the participants. To still capture disease severity, we adjusted for hospitalizations and COPD medication other than NAC. Future studies on patients using NAC, with clinical data on disease severity, are warranted.

Exposure is based on DDD of NAC collected from pharmacies in Sweden. A limitation is that it is not known whether the participants, in addition, have purchased NAC over the counter. It is however unlikely to have a significant impact in a Swedish setting due to subsidization of prescribed medicine.

Overall, the observational study design is associated with limitations that reduces the ability to infer causality; due to the landmark study design there may be survival bias present as participants were required to survive until start of follow-up in 2020. We lack certain information about the participants that would have greatly improved our ability to adjust for confounders, e.g. GOLD severity standard and severity of depression. While we have tried to overcome this limitation by using proxy variables such hospitalization and other COPD medication use, we have a significant risk of confounding by indication as an increased COPD severity may lead to both higher consumption of NAC and increased risk of developing depression.

5. Conclusions

Contrary to our hypothesis our results indicate an increased risk of

developing depression when patients with COPD and/or chronic bronchitis are treated with NAC. This most likely reflects that the consumption of NAC work as a proxy for the severity of disease, where patients with more severe disease collect higher quantities of NAC and also develop depression to a higher degree.

CRediT authorship contribution statement

Anna Englund: Writing – original draft, Formal analysis, Data curation. **Tessa Schillemans:** Writing – review & editing, Methodology. **Klas Ytterbrink Nordenskiöld:** Writing – review & editing, Methodology. **Johanna Wallensten:** Writing – review & editing, Investigation. **Axel C. Carlsson:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Caroline Wachtler:** Writing – review & editing, Supervision, Project administration.

Consent statement

Informed consent was not applicable or required for this study.

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Declaration of competing interest

None to declare.

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None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2026.121905>.

Data availability

The data from this study are available to qualified researchers trained in confidentiality protocols for human subjects, upon obtaining ethical approval from Region Stockholm.

Researchers with appropriate qualifications and training in human subject confidentiality protocols may request access to the data from this study for research purposes, pending ethical approval from Region Stockholm (contact: halsodata.rst@regionstockholm.se). The analytical code and software used in this study are available upon request from axel.carlsson@ki.se.

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